

HW 6

In slides

b-(1) the purple shaded area is a 2 to 1 MUX that selected between "new inputs" and "current outputs" based on the loaded signal, allowing data loading or retention without disturbing the clock signal

b-(2)	x	y	Q	S	Q next	J	k
	0	0	0	0	0	0	x
	0	0	1	1	0	x	1
	0	1	0	1	0	0	x
	0	1	1	0	1	x	0
	1	0	0	1	0	0	x
	1	0	1	0	1	x	0
	1	1	0	0	1	1	x
	1	1	1	1	1	x	0

$$S = x \oplus y \oplus Q$$

$$J = xy$$

$$k = x'y' = (x+y)'$$

b-(3) achieving a "Toggle" function by feeding the $\bar{Q}$ output back to the D input of each flip-flop to simulate binary counting. This setup uses negative-edge triggered flip-flops, where each stage is triggered by the 1 to 0 transition of preceding stage, creating a serial ripple effect for binary sequence

b-(4) The circuit utilizes Jk flip-flop's toggle at $J=k=1$.

Q_8 feedback to prior stages ensures: after 1001, the next pulse resets Q_8 to 0 while steering the ripple to return the register to 0000

b-(5) This is a synchronous counter, b.8(a) is a asynchronous circuit

b-(6) verify the logic of a 4 bit synchronous counter the A_0 always toggles, while a higher bit A_n toggles only when all lower are 0

Problem



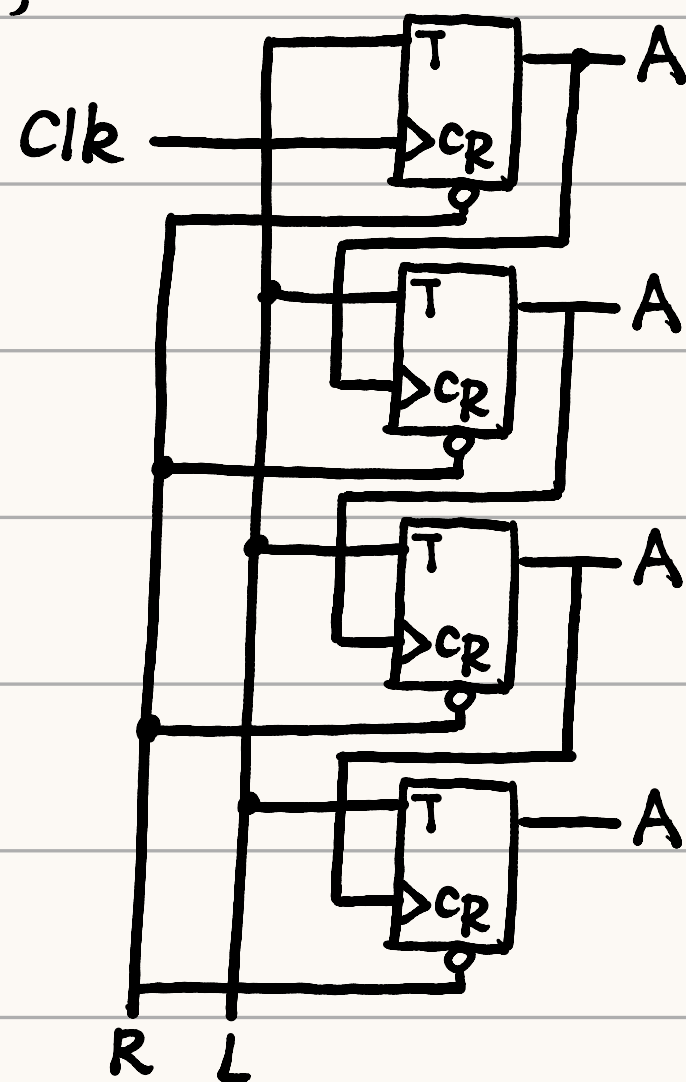
Gating the clock with a NAND gate can causes issues when the load signal is asynchronous: It may induce metastability if the gated clock violates setup time, truncate the clock pulse width and introduce

propagation delay

10111000110

b.4 0110 → 0011 0001 1000 1100 1110 0111

b.12(a)



$$b.21(a) \quad J = L'C + LI \quad k = LI' + L'C$$

b.30 a Johnson counter with n flip-flops feeds the inverted last output back to the first.

00000 → 10000 → 11000 → 11100 → 11110 → 11111 →
01111 → 00111 → 00011 → 00001 → (00000)

$$A_4 A_3 A_2 A_1 A_0 \quad \bar{A}_0 A_4 A_3 A_2 A_1 \quad \bar{A}_1 \bar{A}_0 A_4 A_3 A_2$$

$$\bar{A}_2 \bar{A}_1 \bar{A}_0 A_4 A_3 \quad \bar{A}_3 \bar{A}_2 \bar{A}_1 \bar{A}_0 A_4 \quad \bar{A}_4 \bar{A}_3 \bar{A}_2 \bar{A}_1 \bar{A}_0$$

$$A_0 \bar{A}_4 \bar{A}_3 \bar{A}_2 \bar{A}_1 \quad A_1 A_0 \bar{A}_4 \bar{A}_3 \bar{A}_2 \quad A_2 A_1 A_0 \bar{A}_4 \bar{A}_3$$

$$A_3 A_2 A_1 A_0 \bar{A}_4$$

b.4] The circuit is a serial odd-parity generator using

a D-flip flop and an XOR gate $P(t+1) = D \oplus P$, the

output is cleared by the asynchronous. Since XOR

is a modulo-2 addition, the output toggles for each input

1 and hold 0, so the number of 1 results 1